

Week-1

Tic Tac Toe

Input

o

```

import random

board = [" "] * 9

def show_board():
    print(board[0], "|", board[1], "|", board[2])
    print("-----")
    print(board[3], "|", board[4], "|", board[5])
    print("-----")
    print(board[6], "|", board[7], "|", board[8])

def winner(player):
    wins = [
        (0,1,2), (3,4,5), (6,7,8),
        (0,3,6), (1,4,7), (2,5,8),
        (0,4,8), (2,4,6)
    ]
    for a, b, c in wins:
        if board[a] == board[b] == board[c] == player:
            return True
    return False

while True:

    show_board()
    pos = int(input("Your turn (1-9): ")) - 1

    if board[pos] != " ":
        print("Position already occupied!")
        continue

    board[pos] = "X"

    if winner("X"):
        show_board()
        print("You win!")
        break

    if " " not in board:
        show_board()
        print("It's a draw!")
        break

    empty = [i for i in range(9) if board[i] == " "]
    pos = random.choice(empty)
    board[pos] = "O"

    print("Computer chose:", pos + 1)

    if winner("O"):
        show_board()
        print("Computer wins!")
        break

    if " " not in board:
        show_board()
        print("It's a draw!")
        break

```

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  |  |
--+--+--+
  |  |
--+--+--+
  |  |
Your turn (1-9): 5
Computer chose: 8
  |  |
--+--+--+
  | X |
--+--+--+
  | O |
Your turn (1-9): 1
Computer chose: 3
X |  | O
--+--+--+
  | X |
--+--+--+
  | O |
Your turn (1-9): 9
X |  | O
--+--+--+
  | X |
--+--+--+
  | O | X
You win!

```